

CSE260

Assignment 03 (Total Marks: 25)

This assignment must be hand-written. Show ALL steps in ALL questions.

Make sure that your circuit is efficient, meaning you should use the lowest number of components. You may use external gates if required.

Answer Question 1,2,3,4,5

Question 1 (5 Marks)

Build a circuit that implements the 1's complement number system (3 bit) using encoder(s) and decoder(s).

Question 2: (5 Marks)

Implement the following boolean function using only one 4:1 mux

$$F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$$

Use external gates if required.

Question 3: (5 Marks)

Implement the following boolean function using 2x4 decoder(s) only

$$F(A,B,C,D,E) = \sum(0,1,2,7,8,10,11,13, 15,18,21,24,25)$$

Use external gates if required.

Question 4: (5 Marks)

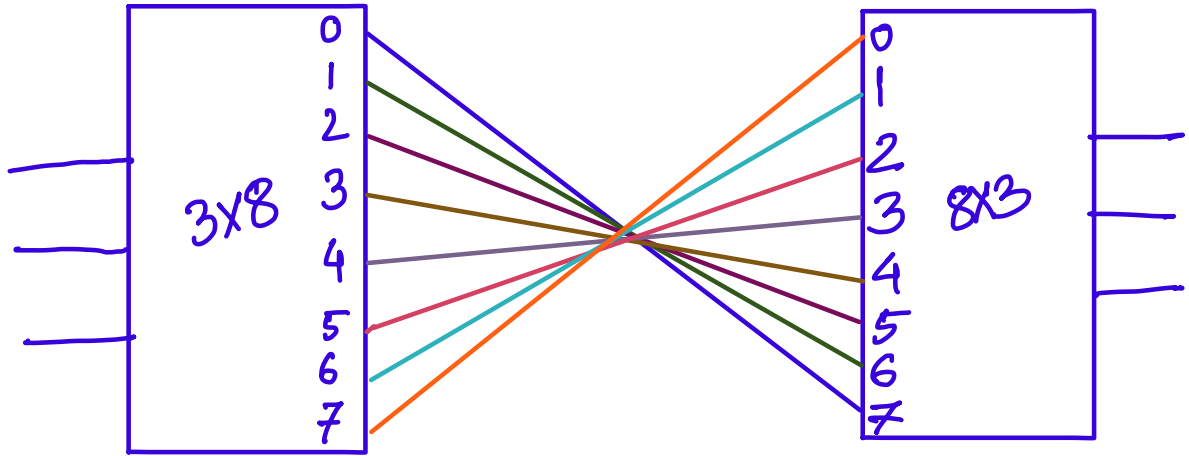
Consider A is a 4 bit number. Design A-3 (A minus 3) using a 4 bit parallel adder. Use external gates if required. The internal diagram for the parallel adder must be drawn using Encoders and decoders.

NB: Inside the parallel adder, the components you see are basic full adders. For a 4 bit parallel adder, you need 4 full adders. Those full adders must be built using encoder(s) and decoder(s).

Question 5 (5 Marks)

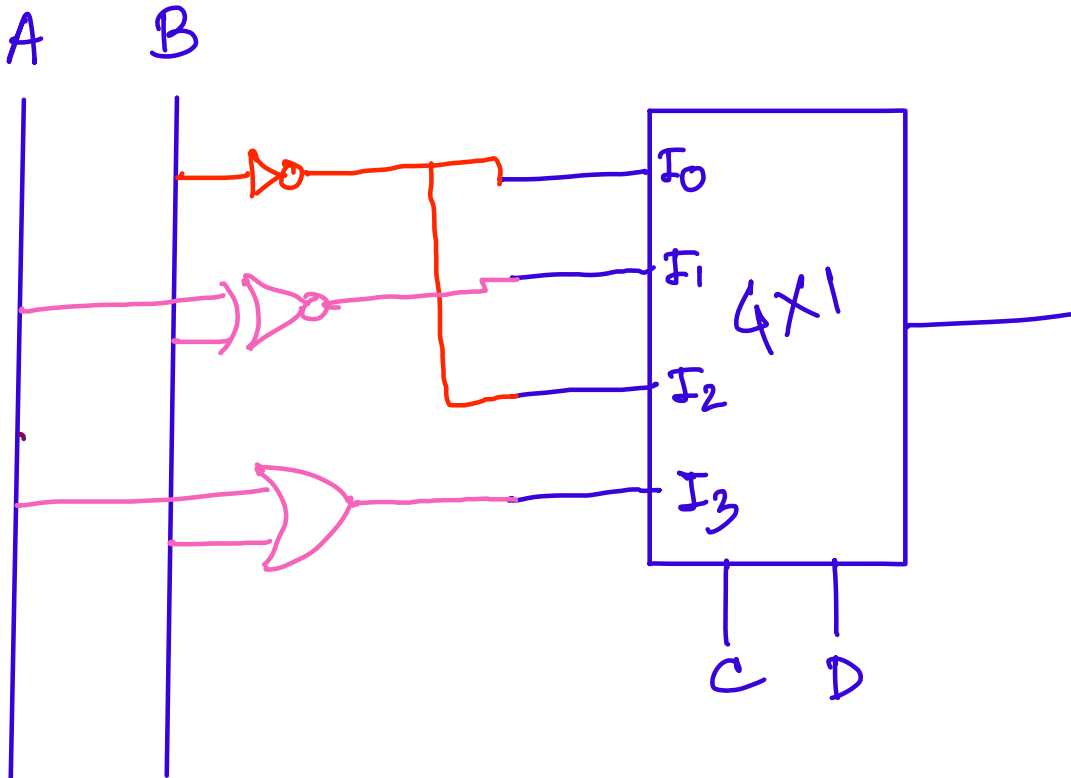
You want to build a calculator that can add or subtract two 4-bit numbers: X and Y. Now, if X is greater than or equal to 8, and Y is divisible by 12, then the calculator will do the operation X+Y (addition). Otherwise, the calculator will do the operation X-Y(subtraction). **Design** the circuit using a 4-bit parallel adder cum subtractor.

1



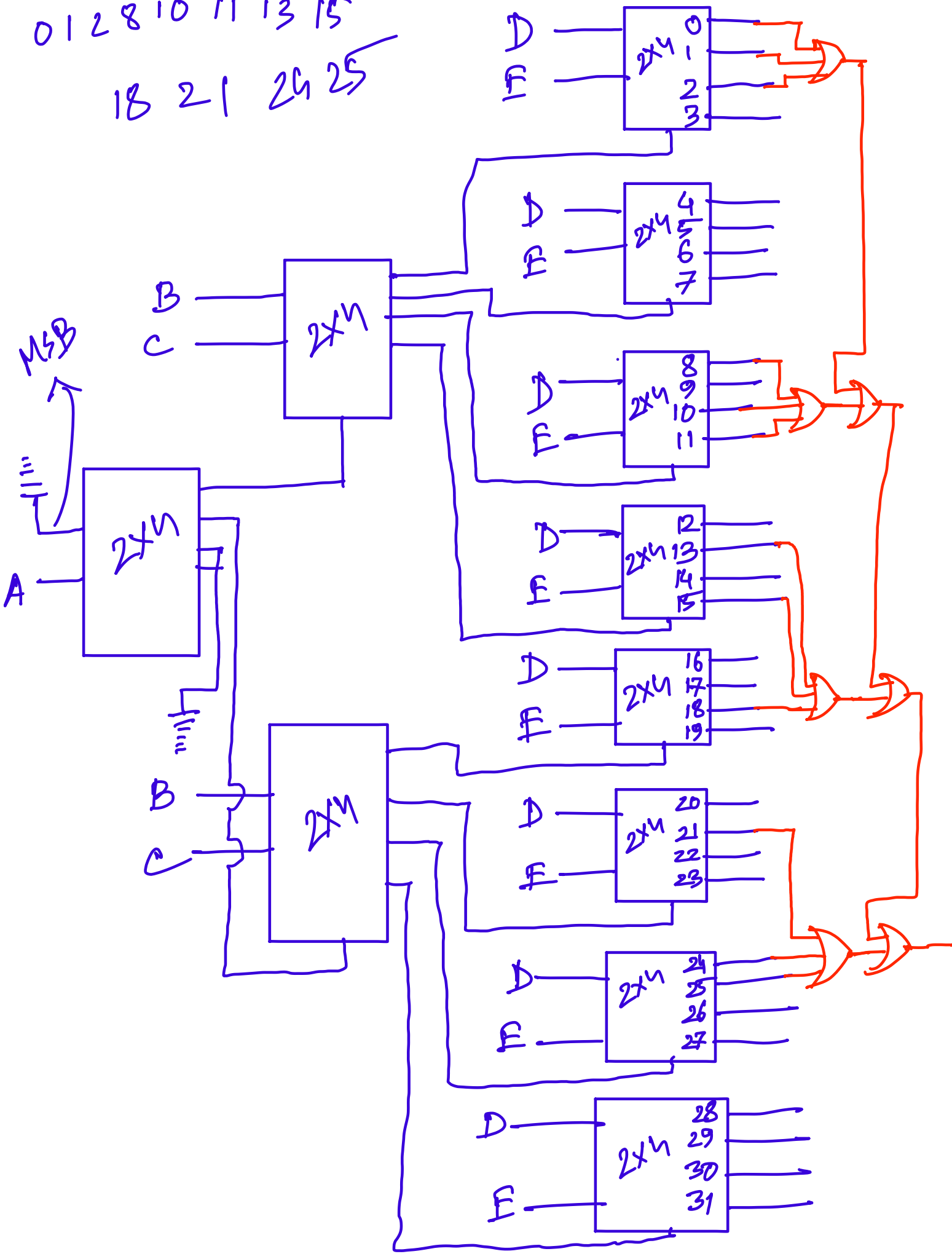
2

	I_0	I_1	I_2	I_3
$A'B'$	(0)	(1)	(2)	3
$A'B$	4	5	6	(7)
AB'	(8)	9	(10)	(11)
AB	12	(13)	14	(15)
	B'	$A \odot B$	B'	$A+B$



3

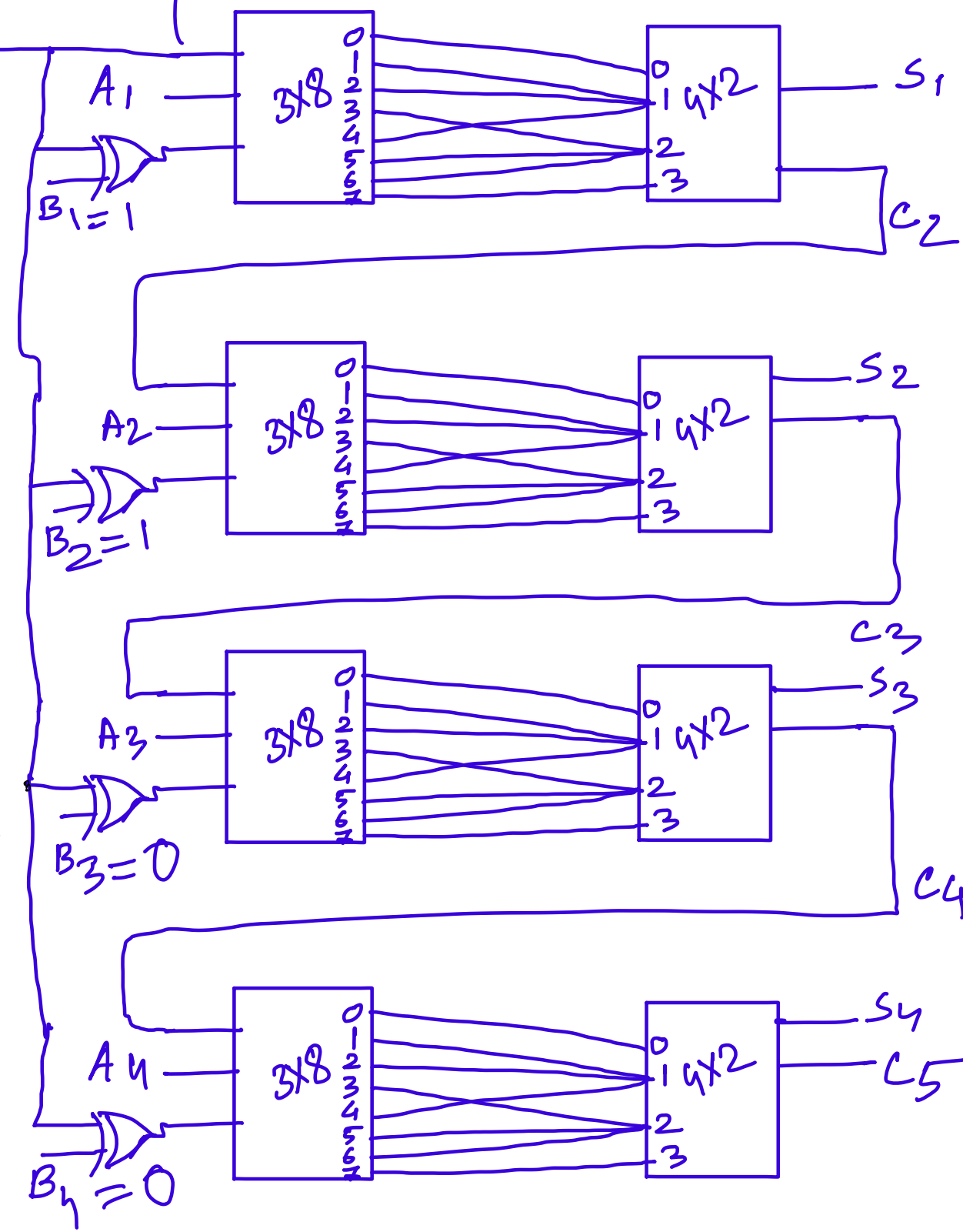
0 1 2 8 10 11 13 15
18 21 24 25



$00 \rightarrow 0$
 $01 \rightarrow 1, 2, 4$
 $10 \rightarrow 3, 5, 6$
 $11 \rightarrow 7$

4

$c_{in} = 1$



5

X is greater than or equal to 8 $\rightarrow$ MSB 1

Y is divisible by 12 $\rightarrow$ 0, 12

0000

1100

$$Y_4'Y_3'Y_2'Y_1' + Y_4Y_3Y_2'Y_1'$$

